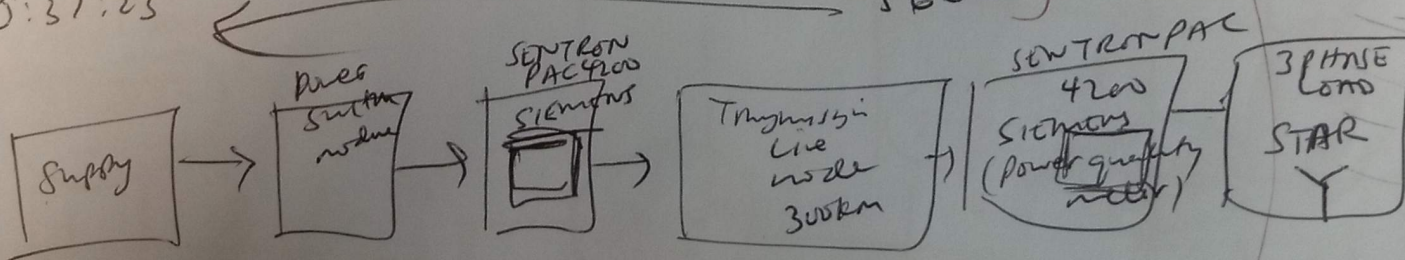
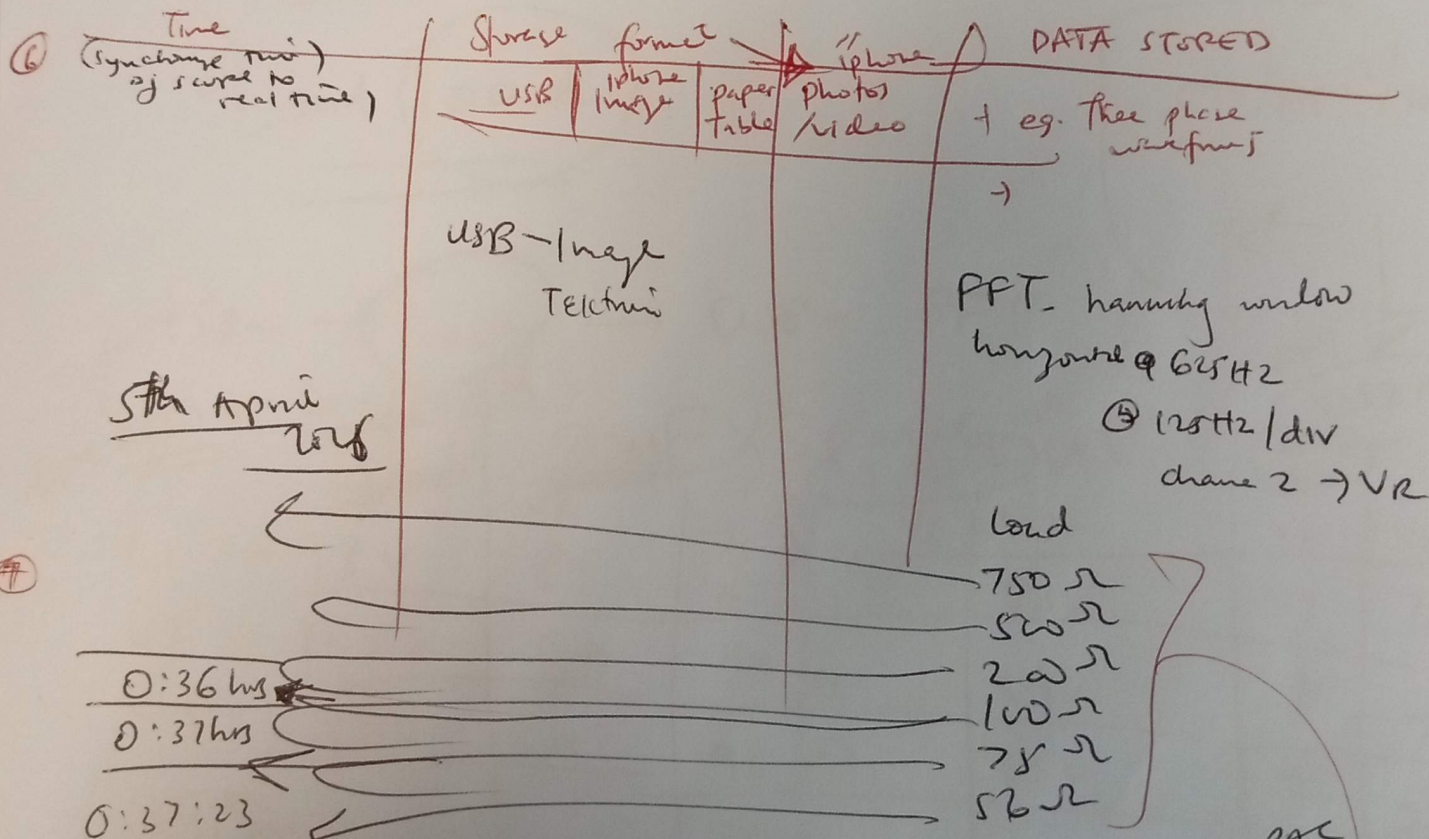


RESULTS

- Frequency sweep from 0Hz - 60Hz $\xrightarrow{50\%}$ 120Hz $\rightarrow$ π model - SSO
- 11 11 ~~120Hz~~ $\rightarrow$ 1mHz $\rightarrow$ π model - Super Synchronous oscillations & other harmonic content.

Systematic results

- * $\rightarrow$ START Tablig / storing / capturing
- 1 $\rightarrow$ Calibrate instrument / Adjust settings to get desired output signal.
- 2 Check if signal is correct both in magnitude & shape
- 3 Compare with expected output result / waveform.
- 4 Feed into the physical model (signal)
- 5 Compare input vs output.
- 6 TAKE PICTS OF SETUP (EXPERIMENT)



Supply 5MVA/10kV

* 0:55 AM $\rightarrow$ iPhone Image $\rightarrow$ SETUP.

* 1:00 AM $\rightarrow$ //

Waveforms @ load - Resistive

- ② Oscilloscope
- ③ Voltages (V_S & V_R)
- LN

AN EXPERIMENT SHOULD BE REPLICABLE.

	V_s	V_R	P_{Watts}	Amps Instantaneous
L1	99V	77V	55 Watts	0.72A
L2	101V	76V	54 Watts	0.72A
L3	97V	79V	56 Watts	0.70A
			Neutral	0.01A

$$L = 230mH \times 3 \text{ Lines}$$

$$GND = 15\Omega + 200mH$$

Load 100 Ω

5th/Apr/2026

2:45 AM 100 Ω $\rightarrow$ USB - FFT
 FFT 2:51 AM Capacitor
SERIES 30M $\times$ 3 phase USB
 2:51:56 FFT

	V_s	V_R		V_s	V_R		V_s	V_R
L1	98V	89	75	83V			97	73
L2	100V	89	99	83V			99	74
L3	96V	93	96	85V			95	75

FFT 2:51 AM 200 Ω Y load

	V_s	V_R
L1	99	95
L2	98	96
L3	98	98

50 Ω

	V_s	V_R
L1	100	100
L2	103	100
L3	99	102

USB FFT
2:50:47

750 Ω

	V_s	V_R
L1	101	102
L2	104	103
L3	100	105

USB FFT
2:50

30 MF / 450V

SERIES Compensation 3φ

750Ω → USB Tektronix → time
 (WAVE FORM ON V_s as V_R) 2:55 AM

500Ω // // → 2:55:58 AM

200Ω // // → 2:56:33 AM

100Ω // // → 2:56:51 AM

75Ω // // → 2:57:07 AM

56Ω // // → 2:57:21 AM

5th
 April
 2026

4μF

100Ω load → USB waveforms V_R & V_s
 3:13:17 AM

75Ω // // 3:13:50 AM

56Ω // // 3:14: AM

4μF

750Ω load

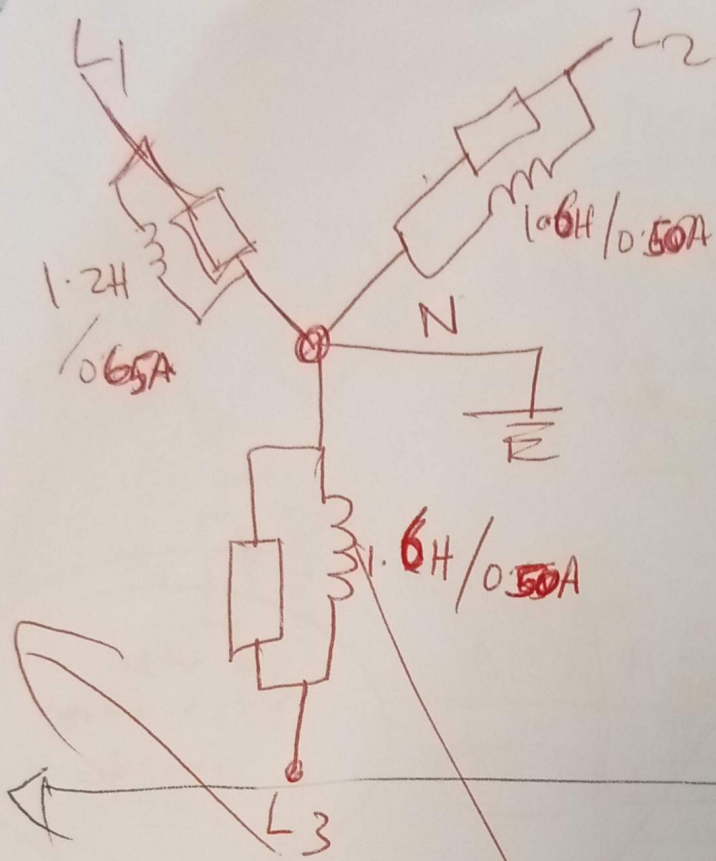
	V_s	V_R
21	102	76
62	101	77
13	100	78

SAME SETUP

Load (image) - phone

3:26 AM - 5th/

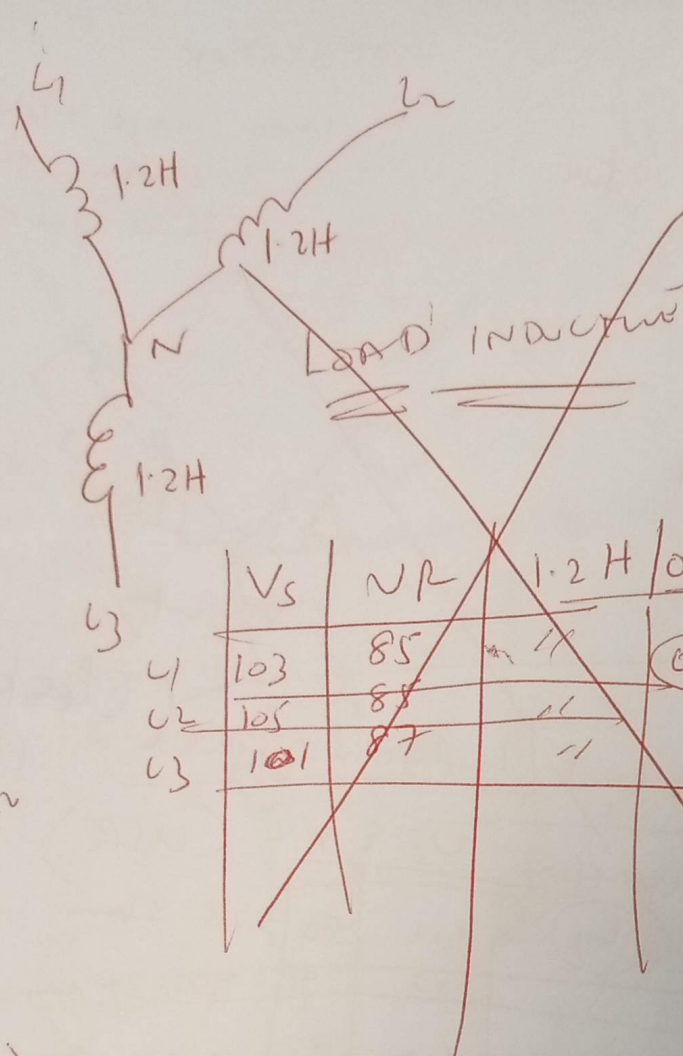
April
2012



Variable ohmic load / Inductor Constant

(clock left)

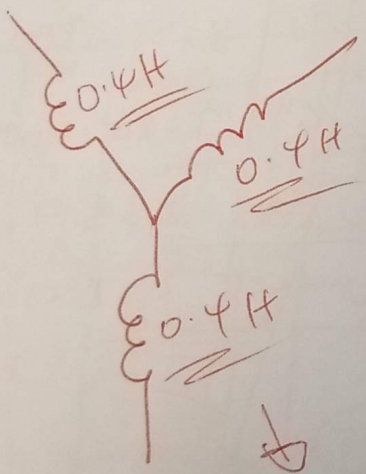
	Wrefm	V_s	V_R	(USB)
750 Ω	(L1)	102	80	3:28:33 AM
500 Ω		101	79	3:29:39 AM
200 Ω		100	75	3:29:58 AM
100 Ω		99V	67V	3:30:20 AM
75 Ω		99V	59V	3:30:42 AM
50 Ω		99V	50V	3:31:05 AM



	V_s	NR	1.2H	0.65A
u_1	103	85	11	u_1 0.23A
u_2	105	88	11	
u_3	101	87	11	

USB
3:37:47 AM
IMAGE

Corrected
in the
next
pages.



10mH at 120Hz

3:46:22 AM

Abandon results -

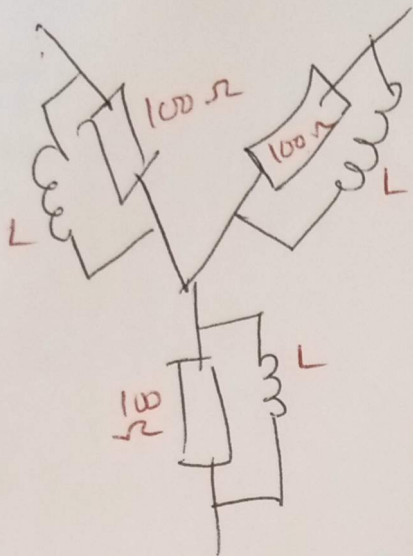
$3.2H \rightarrow 2.4H \rightarrow 0.8H$ 3:59:15 AM
 $3.2H \rightarrow 2.0H \rightarrow 1.2H$ 3:59:47 AM
 $3.2H \rightarrow 1.6H \rightarrow 1.6H$ 4:00:18 AM
 $3.2H \rightarrow 1.2H \rightarrow 2H$ 4:01:46 AM
 $\rightarrow 3.2H \rightarrow 3.2H$ 4:02:54 AM

Old
 Similar
 waveform
 thereafter
 until
 4:06 AM

$2 \cdot 2H = 3.2$
 $0.01944 = 3.2$
 $0.01 + 3.2$

$2 \cdot 3.2$
 $\frac{2.8}{0.4H}$
 $=$

100 Ω Constant



✓ Phase to neutral

$L = 1.2H$

	V_S	V_R
L1	100	66
L2	102	67
L3	98	69

16:25

FFT 16:31:21

16:32

16:32:25 $\rightarrow$ 80Hz 350Hz

16:33:04 $\rightarrow$ 400Hz

$L = 1.6H$

	V_S	V_R
L1	100	68
L2	101	69
L3	97	71

16:34:38

16:38:20 FFT $\rightarrow$ 16:35:27 $\rightarrow$ 16:35:41

16:36:00 - 300Hz 400Hz

- Harmonics
- 50Hz fundamental
 - 150Hz
 - 250Hz
 - 350Hz
 - 450Hz
 - 550Hz
 - 650Hz
 - 750Hz - 16:37

2.8H stopped

$L = 2.0H$

	V_S	V_R
L1	99	69
L2	101	70
L3	97	72

Wave form $\rightarrow$ 16:40:02

FFT $\rightarrow$ 16:40:27
 16:40:44
 16:41:03
 16:41:31 - 400Hz

$L = 2.4H$

	V_S	V_R
L1	99	70
L2	101	71
L3	97	73

when V_S & V_R 17:28:49

FFT - 17:28:07

3.2H

	V_S	V_R
L1	99	71
L2	100	72
L3	96	74

17:30:20

17:30:46 - FFT

75 Ω Constant

1.2H

	V_S	V_R
L1	100	59
L2	103	59
L3	99	62

wave
form

17:35:34

FFT $\rightarrow$ 17:37:15

2.8H

	V_S	V_R
L1	101	63
L2	103	62
L3	99	66

FFT $\rightarrow$ 17:42:32
17:42:54 - waveform

1.6H

	V_S	V_R
L1	100	60
L2	102	60
L3	98	63

17:38:39 - FFT

17:39:10 - waveform
 V_S & V_R

3.2H

	V_S	V_R
L1	101	63
L2	103	63
L3	99	66

17:44:09 - waveform
17:44:30 - FFT

2.0H

	V_S	V_R
L1	101	61
L2	103	61
L3	99	64

wave
form $\rightarrow$ 17:40:37

FFT - 17:41:14

56 Ω constant

V_{ph}-neutral instantaneous

2.0H / 0.45A

1.2H / 0.65A

1.6H / 0.50A

	V _S	V _R
L ₁	101	50
L ₂	103	51
L ₃	99	53

	V _S	V _R
L ₁	100	51
L ₂	103	52
L ₃	99	54

	V _S	V _R
L ₁	100	52
L ₂	103	53
L ₃	99	55

17:46:02 → waveform
V_S & V_R
17:46:25 → FFT

17:47-FFT
17:48:01 → waveform
V_S & V_R

waveform
17:49:00
17:50:27
↓
FFT

1.6H

2.4H / 0.35A

2.8H / 0.30A

3.2H / 0.25A

	V _S	V _R
L ₁	100	52
L ₂	102	53
L ₃	98	55

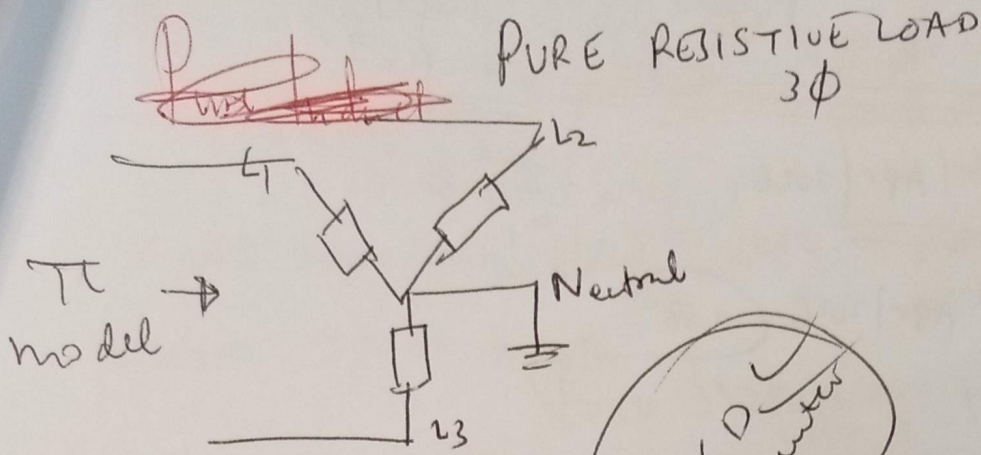
	V _S	V _R
L ₁	100	52
L ₂	102	53
L ₃	98	55

	V _S	V _R
L ₁	100	53
L ₂	102	54
L ₃	98	56

17:51:26 - FFT
17:51:44 - waveform

17:52:57 → waveform
17:53:15 → FFT

17:54:53
↓
FFT
17:55:12
↓
waveform
(V_S & V_R)



$V_{pn-neutral}$

ABCD
resistor

DATE
5 APR 2026

18:06:13 $\rightarrow$ FFT with 250Hz cursor
150Hz $\leftarrow$ FFT $\leftarrow$ 18:05:41
18:05:06 waveform V3 & VR

750 Ω			500 Ω		
L1	V_S	V_R	L1	V_S	V_R
L1	102	101	L1	101	97
L2	104	102	L2	103	98
L3	100	105	L3	99	100

18:08:00 - waveform V3 & VR
18:08:06 - FFT - 106 Hz shown cursor
18:08:25 - FFT - 250 Hz shown cursor

18:02:19 - FFT
18:03:43 - waveform V3 & VR
Yellow - sending end
Blue - receiving end

200 Ω

L1	V_S	V_R
L1	101	87
L2	102	88
L3	98	91

75 Ω

100 Ω			75 Ω		
L1	V_S	V_R	L1	V_S	V_R
L1	99	76	L1	99	66
L2	101	75	L2	100	66
L3	97	78	L3	97	67

FFT $\rightarrow$ 200 Ω $\rightarrow$ 18:07:30
18:07:48 $\rightarrow$ FFT with 150Hz cursor
18:08:31 - waveform

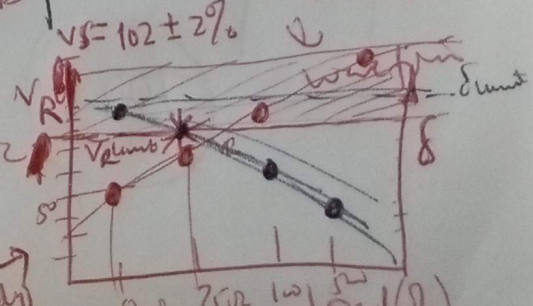
56 Ω

L1	V_S	V_R
L1	98	53
L2	100	54
L3	97	56

18:15:13 $\rightarrow$ FFT
18:15:43 $\rightarrow$ FFT
18:16:07

18:10:19 - waveform V3 & VR
18:10:44

18:11:05
FFT with 400Hz



DLR - dynamic line rating
Extra current $\rightarrow$ voltage drop
Realized along the line
 $Z = R + j\omega L$

56Ω → 3phases V_R and V_S phase shift waveforms.

$L_1 \rightarrow 18:17:41$ 5th Apr/2026 (-55°) → 2 → 1 phase
closed
 V_R V_S

$L_2 \rightarrow 18:19:50$ 5th Apr/2026 (-58°) // //

$L_3 \rightarrow 18:20:42$ //

	V_S	V_R
L_1	98V	53V
L_2	100V	54V
L_3	97V	56V

~~PHASE SHIFT WAVEFORMS~~
For ALL THREE LINES L_1, L_2 & L_3

100Ω

	V_S	V_R
L_1	99	76
L_2	101	75
L_3	97	79

closed channel
2 → 1 phase

$V_R \rightarrow V_S$ phase shift

$L_1 \rightarrow 18:21:32 \rightarrow 38^\circ$

$L_2 \rightarrow 18:23:57 \rightarrow 38^\circ$

$L_3 \rightarrow 18:22:49 \rightarrow 37^\circ$ (value)

500Ω

	V_S	V_R
L_1	100	97
L_2	103	97
L_3	99	99

$V_R \rightarrow V_S$

$L_1 \rightarrow 18:28:33 \rightarrow -14^\circ$ min

$L_2 \rightarrow 18:30:14 \rightarrow -14^\circ$

$L_3 \rightarrow 18:31:15 \rightarrow -15^\circ$

700Ω

	V_S	V_R
L_1	101	101
L_2	104	102
L_3	100	104

$L_1 \rightarrow 18:36:37$ (8.50)

$L_2 \rightarrow 18:34:27$ (8.50)

$L_3 \rightarrow 18:33:01$ (-8.50)

310
250

Some where between 500-600 Hz

The 150 Hz harmonic decays rapidly, why?

Supply $\rightarrow$ π model $\rightarrow$ load 3ϕ Y load
 3ϕ -YN LN (500-600 Hz)

At high load (50 Hz) - 150 Hz is prominent (3rd in magnitude) why?

a near 250 Hz $\rightarrow$ both 150 Hz & 250 Hz are almost same magnitude why?

350 Hz - ~~decays~~ decays with increase in load power (750 $\rightarrow$ 50 Hz) why?

480 Hz $\rightarrow$ grows with increase in load power (750 $\rightarrow$ 50 Hz) ?
Frequency = 50.08 Hz

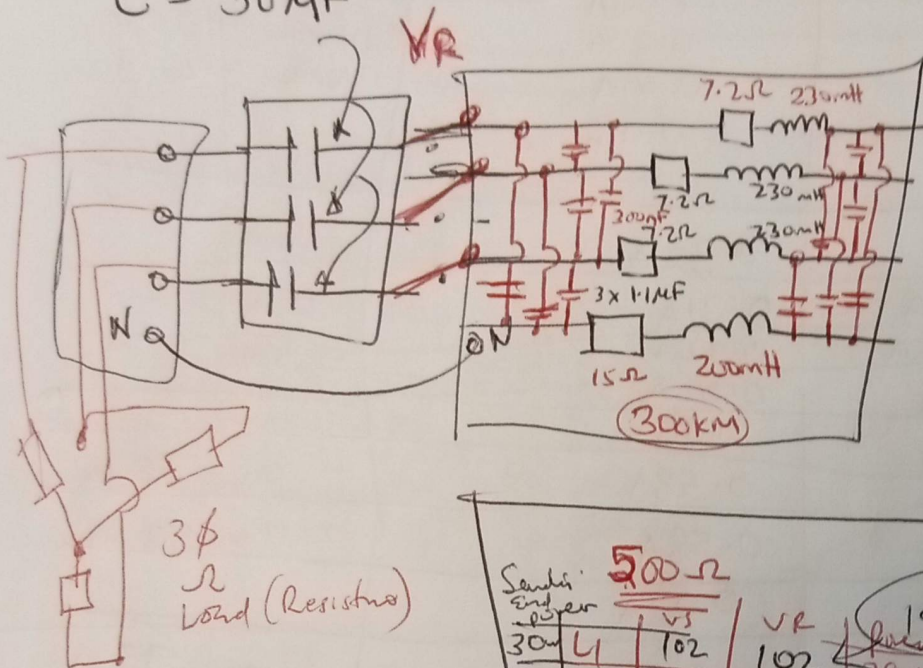
	V_s	Power (Watt)	Amps	V_R	Phase unbalance % $V_{nb} \rightarrow$ % $I_{nb} \rightarrow$	THD I(A) $L_1 \rightarrow$ $L_2 \rightarrow$ $L_3 \rightarrow$	THD V_{ph-n} $L_1 \rightarrow$ $L_2 \rightarrow$ $L_3 \rightarrow$
56 Ω	$L_1 \rightarrow 99$ $L_2 \rightarrow 101$ $L_3 \rightarrow 98$	55 55 57	1.02 A 1.01 A 1.01 A N $\rightarrow 0.03$	54 55 57	1.0% 0.6%	1.3% 1.1% 0.7%	9.1% 6.3% 0.9%
75 Ω	$L_1 \rightarrow 99$ $L_2 \rightarrow 101$ $L_3 \rightarrow 97$	65 64 67	0.89 A 0.88 A 0.87 A N $\rightarrow 0.02$	61 62 63 59		5.6% 4.1% 1.0%	1.1% 0.9% 0.9%
100 Ω	$L_1 \rightarrow 99$ $L_2 \rightarrow 101$ $L_3 \rightarrow 98$	75 76 79	0.73 A 0.72 A 0.72 A N $\rightarrow 0.02$	71 72 73 69		3.7% 2.7% 1.4%	1.4% 1.2% 1.3%
200 Ω	$L_1 \rightarrow 101$ $L_2 \rightarrow 103$ $L_3 \rightarrow 99$	89 89 92	0.49 A 0.50 A 0.49 A N $\rightarrow 0.01$	84 85 86 82		2.6% 2.3% 1.9%	1.8% 1.7% 1.9%
500 Ω	$L_1 \rightarrow 102$ $L_2 \rightarrow 104$ $L_3 \rightarrow 107$	98 99 101	0.27 A 0.27 A 0.27 A N $\rightarrow 0.01$	93 94 95 91		1.9% 2.0% 2.0%	1.8% 1.8% 2.0%
750 Ω	$L_1 \rightarrow 103$ $L_2 \rightarrow 105$ $L_3 \rightarrow 101$	102 103 106	0.14 A 0.14 A 0.14 A N $\rightarrow 0.00$	98 99 100 96		2.3% 2.4% 2.3%	2.5% 2.4% 2.3%

SERIES CAPACITOR COMPENSATION

5th APR 2026

SETUP → 19:36 → 37 PM
(phone images)

$C = 30 \mu F$



750 Ω

Power		V_s	V_R	Watts
16w	L1	103	104	15w
16w	L2	105	104	15w
15w	L3	102	107	15w

19:46:40 → waveform
(V_s & V_R)
19:47:14
19:47:32 PFT
19:48:15 200Hz

Angle
Inductor
VR
Voltage
Stability
Series
Compensation

500 Ω

Power		V_s	V_R	Watts
30w	L1	102	102	28w
30w	L2	104	102	28w
30w	L3	101	104	29w

19:51:29
PFT
19:52:57 PFT → 200Hz
19:52:57 PFT → 300Hz
19:52:57 WAVEFORM (V_s & V_R voltage angle)

200 Ω

Power		V_s	V_R	Watts
54w	L1	101	97	50w
53w	L2	103	97	48w
54w	L3	100	100	51w

100 Ω

Power		V_s	V_R	Watts
88w	L1	100	92	78w
88w	L2	102	91	77w
87w	L3	98	94	80w

75 Ω

Power		V_s	V_R	Watts
111w	L1	99	85	96
110w	L2	101	85	95
110w	L3	98	86	96

56 Ω

Power		V_s	V_R	Watts
126w	L1	99	75	103w
126w	L2	100	76	105w
124w	L3	97	77	106w

$\phi = 12^\circ$
20:00:10 - waveform
20:00:38 - PFT
20:00:53 - PFT
20:01:20 - PFT - 200Hz

150Hz harmonic grows larger, why?

Continuation

Load (Resistive)	Receiving end of TC model		Sending end of TC model		Sending end	
	Ampr	Instantaneous	Ampr	Instantaneous	VAR	VA
750 Ω	L1	0.14A		0.21A	14VAR	21VA
	L2	0.14A		0.20A	14VAR	21VA
	L3	0.14A		0.20A	13VAR	21VA
	N	0.00A		0.01A N		
500 Ω	L1	0.27A		0.32A	15VAR	33VA
	L2	0.27A		0.31A	15VAR	33VA
	L3	0.28A		0.32A	15VAR	33VA
	N	0.01A		0.02A N		
200 Ω	L1	0.54A		0.60A	18VAR	58VA
	L2	0.54A		0.58A	17VAR	56VA
	L3	0.53A		0.58A	17VAR	58VA
	N	0.01A		0.03A N		
100 Ω	L1	0.85A		0.94A	32VAR	93VA
	L2	0.86A		0.91A	29VAR	92VA
	L3	0.85A		0.92A	29VAR	91VA
	N	0.02A		0.05A N		
75 Ω	L1	1.14A		1.24A	51VAR	122VA
	L2	1.13A		1.20A	48VAR	120VA
	L3	1.13A		1.23A	47VAR	118VA
	N	0.05A		0.08A N		
56 Ω	L1	1.40A		1.52A	82VAR	151VA
	L2	1.40A		1.48A	79VAR	149VA
	L3	1.38A		1.50A	76VAR	147VA
	N	0.09A		0.11A N		

8_μF SERIES 3 ϕ COMPENSATION

750 Ω

	V_S	V_R
L1	102	96V
L2	104	97V
L3	101	99V

20:47:05
waveform

(50 Hz full wave
250 Hz - Harmonic (2nd in magnitude)
20:48:30 (5th Apr 2026)

	$V_{R\text{End}}$ watts	$V_{R\text{Amps}}$	$V_{S\text{Amps}}$
L1	13W	0.13A	0.23A
L2	13W	0.14A	0.22A
L3	13W	0.13A	0.23A
N		0.00A	0.01A

All quantities

500 Ω

	V_S	V_R	$V_{R\text{End}}$ watts	$V_{R\text{Amps}}$	$V_{S\text{Amps}}$
L1	102	81V	18W	0.22A	0.34A
L2	104	81V	18W	0.22A	0.32A
L3	101	82V	18W	0.22A	0.33A
N				0.01A	0.02A

150 Hz harmonic of 250 &
same magnitude 20:52:24

20:51:14 $\rightarrow$ waveform
PFT
150 Hz grows bigger w/

200 Ω

	V_S	V_R	$V_{R\text{End}}$ watts	$V_{R\text{Amps}}$	$V_{S\text{Amps}}$
L1	103	57V	17W	0.30A	0.48A
L2	105	58V	17W	0.31A	0.48A
L3	102	57V	17W	0.30A	0.48A
N				0.01A	0.02A

$\phi = 50^\circ$

20:54:37 $\rightarrow$ waveform

100 Ω

	V_S	V_R	$V_{R\text{End}}$ watts
L1	103	36V	12W
L2	105	36V	13W
L3	102	37V	12W

$\phi = 60^\circ$

20:56:32 waveform

20:56:57 - PFT

	$V_{R\text{Amps}}$	$V_{S\text{Amps}}$
L1	0.33A	0.48A
L2	0.34A	0.48A
L3	0.34A	0.48A
N	0.02A	0.01A

75 Ω

	V_S	V_R	$V_{R\text{Amps}}$	$V_{S\text{Amps}}$
L1	103	26V	0.49A	0.34A
L2	105	27V	0.49A	0.35A
L3	102	27V	0.49A	0.35A
N			0.02A	0.01A

20:57:38 - PFT

$\phi = 63^\circ$

20:58:30 $\rightarrow$ waveform

$V_{R\text{End}}$
watts
L1 - 9W
L2 - 9W
L3 - 9W

56 Ω

$\phi = 66^\circ$

	V_S	V_R	$V_{R\text{Amps}}$	$V_{S\text{Amps}}$
L1	103	19V	0.5A	0.35A
L2	106	20V	0.5A	0.36A
L3	102	20V	0.5A	0.35A
N			0.02A	0.01A

20:59:27 - waveform
20:59:58 PFT

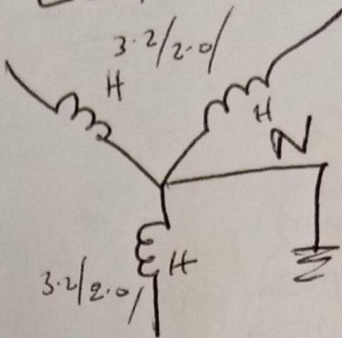
8uF Compensation

Pure resistive (Inductive)
Load

30uF Inductive

Load
3.2H

	Vs	VR	Vs Amps	VR Amps
L1	52	75	0.02	0.08A
L2	53	74	0.02	0.08A
L3	50	76	0.02	0.08A
N			0.01	0.08A



2.0H

	Vs	VR	Vs Amps	VR Amps	VA		Watts		VAR	
					Vs	VR	Vs	VR	Vs	VR
L1	52	99V	0.11A	0.17A	6VA	17VA	2w	2w	5var	16var
L2	53	96V	0.10A	0.16A	5VA	15VA	2w	2w	5var	15var
L3	50	100V	0.11A	0.16A	6VA	16VA	2w	2w	5var	16var
N			0.07A	0.01A						

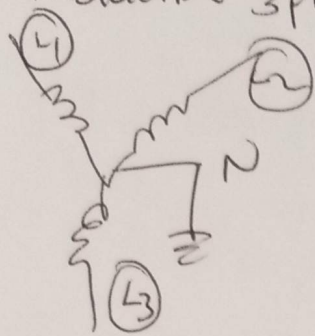
when 22=08
for 22=09 FFP

22:17

1.2H

	Vs	VR	Current I (Amps)		Apparent Power (VA)		Power (watts)			
			Vs Instruments	VR	Vs	VR	VR	Vs	VARs	Vs / VR
L1	33	147	0.41A	0.42A	14VA	62VA	9W	10W	10	63
L2	34	132	0.34A	0.36A	11VA	47VA	6W	8W	9	48
L3	36	137	0.37A	0.38A	11VA	52VA	7W	9W	7	53
N			0.04A	0.02A						

30MF series Compensation Inductive 3phase Load



1.2 H

	V_S	V_R
L1	102	110
L2	104	110
L3	101	113

→ 22:30 waveform
22:30:45 - PF

2.0 H

22:31:16 - PF
22:31:35 - waveform

	V_S	V_R
L1	103	107v
L2	105	107v
L3	102	110v

2.8 H

22:32:13 - waveform
22:32:49 PF

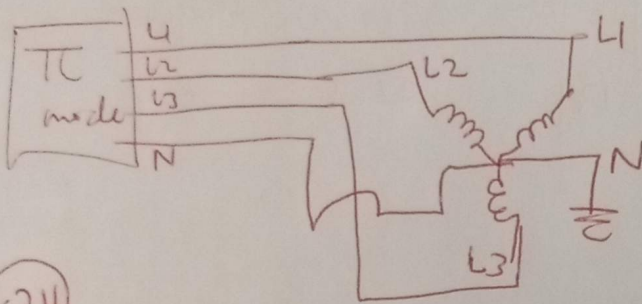
	V		I (Amps)		VA		Power (Watts)	
	V_S	V_R	V_S	V_R	V_S	V_R	V_S	V_R
L1	103	107	0.03A	0.13A	3 VA	14 VA	2w	2w
L2	105	107	0.03A	0.12A	3 VA	13 VA	2w	2w
L3	102	110	0.03A	0.13A	3 VA	14 VA	2w	2w
N			0.02A	0.00A				

Total PF Instantaneous

$V_R \text{ End} \rightarrow \underline{0.13}$

$V_S \text{ End} \rightarrow \underline{0.64}$

Inductive Load WITHOUT COMPENSATION



1.2H

	V_S	V_R
L1	102	85
L2	104	86
L3	101	87

22:42 - FFT
22:42 - waveform

1.6H

	V_S	V_R
L1	102	88v
L2	104	89v
L3	101	91v

22:43:24 - waveform
22:43:48 - FFT

2.8H

	V_S	V_R
L1	103	95v
L2	105	96v
L3	102	98v

22:45:00 - FFT
22:45:22 - waveform

Unloaded line without compensation

	V_S	V_R	Amps	
			V_S	V_R
L1	103v	106v	0.14A	0.00A
L2	105v	107v	0.13A	0.00A
L3	102v	109v	0.13A	0.0A
N			0.02A	0.0A

Ferranti Effect

22:48 - waveform
22:48:31 - FFT

VA

	V_S	V_R
L1	14VA	0VA
L2	14VA	0VA
L3	14VA	0VA

WATTS

	V_S	V_R
L1	0W	0W
L2	0W	0W
L3	0W	0W

VARs

	V_S	V_R
L1	14	0
L2	14	0
L3	14	0

SERIES
compensated (30mF) BUT unbalanced

	V_S	V_R	Amps	
L_1	103v	106v	I_{V_S}	V_R
L_2	105v	107v	0.14A	0.0
L_3	102v	109v	0.13A	0.0
			0.13A	0.0
			0.02A	0.0

	V_S	V_R
L_1	14v	0v
L_2	14v	0v
L_3	14v	0v

WATTS

V_S	V_R
L_1	0w 0w
L_2	0w 0w
L_3	0w 0w

	V_S	V_R
L_1	14	0
L_2	14	0
L_3	13	0

PF = 0.01 Sensing End

49.97 Hz

THD Vph-n

V_S	V_R
L_1	2.9% 7.8%
L_2	2.5% 5.8%
L_3	2.7% 7.0%

CAPACITIVE LOAD/UNCOMPENSATED LINE (SERIES)

3 PHASE
30mF
LOAD

23:18:47 - FFT

	V_S	V_R
L_1	31v	108v
L_2	31v	109v
L_3	28v	109v

Amps	V_S	V_R
L_1	1.14	1.05A
L_2	1.14	1.08A
L_3	1.13	1.04A
N	0.07	0.04A

VA	V_S	V_R
L_1	35	113
L_2	35	117
L_3	31	113

Watts	V_S	V_R
L_1	11	0
L_2	12	1
L_3	11	0

	V_S	V_R
L_1	33	112
L_2	33	116
L_3	29	112

THD Vph-n Inst	V_R	V_S
L_1	5.5%	3.1%
L_2	4.2%	2.6%
L_3	0.2%	2.9%

23:28:35 $\rightarrow$ 56 Ω 3 ϕ Load
~~unbalanced~~ Uncompensated Line

L1 - 101 - 56v

L2 - 103 - 56v

L3 - 101 - 59v

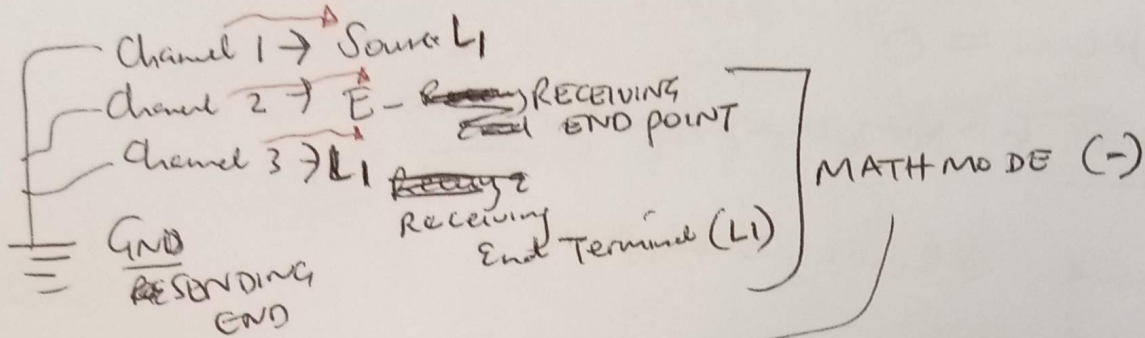
23:29:39 - PPT

$$\delta = 52^\circ$$

SET UP 1100AM 6th/Apr/2026 → iPhone

PI model frequency sweep

0 - 300Hz



V_{L1} VS MATH MODE

3.3V

✶ Square wave Sweep

01:14 - 01:15 AM 6th Apr 2026

* VIDEO - 1:16 - 6th Apr 2026 - iPhone (L1) only

1:20 → (L2) only (L2)

1:22 → (L3) only → (L3) 1:24 AM - USB

SENDING

RECEIVING

L1 - L2 → 1:27 AM - iPhone

L1 → L3 → 1:29 AM iPhone

L2 - L1 → 1:32 AM

L2 - L3 → 1:34 AM

L3 - L1 → 1:36 AM

L3 - L2 → 1:38 AM

SINE WAVE

1:40 AM - iPhone - 2.33 V_{RMS}

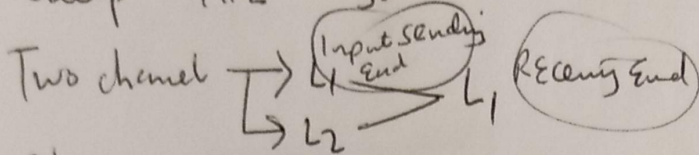
1:42 AM → " → 3.3 V_{RMS}

Sending and Receiving
L1 - L2 end

SINE WAVE 2 Lines, Same phase

Amplitude of signal input - 3.3 Vrms

Swamp - 1Hz - 300Hz



Phase difference = 0°

1:52 AM 6th Apr 2026 $\rightarrow$ Iphone - VIDEO

Square wave 2 Lines, Same phase

$\downarrow$
1:56 AM - Iphone

3 phase Square input 3.3 Vrms

ESP32

$\begin{matrix} L_1 \\ L_2 \\ L_3 \end{matrix} \rightarrow L_1 \text{ measured} \rightarrow$ Iphone 2:29 AM
2:00 $\rightarrow$ 2:30 AM - USB Images

2:33:25 - USB - $\begin{matrix} L_1 \\ L_2 \\ L_3 \end{matrix} \rightarrow L_2 \text{ measured}$ } same frequency point

2:33:57 USB $\begin{matrix} L_1 \\ L_2 \\ L_3 \end{matrix} \rightarrow L_3 \text{ measured}$

2:37 - USB $\rightarrow$ $\begin{matrix} L_1 \\ L_2 \\ L_3 \end{matrix} \rightarrow L_3 \rightarrow$ Amplitude 2nd peak point

2:39 $\rightarrow$ Iphone video $\rightarrow$ $\begin{matrix} L_1 \\ L_2 \\ L_3 \end{matrix} \rightarrow L_3 \text{ measured}$

All phase are uniform

* 2:53:10 AM $\rightarrow$ USB $\rightarrow$ 3 ϕ Receiving End wave form $\begin{pmatrix} L_1 \\ L_2 \\ L_3 \end{pmatrix}$
* 2:58 AM - Iphone VIDEO

STRIP
Compensated line
 3 ϕ Square wave sweep (ESP32)
 phase to phase 120° shift

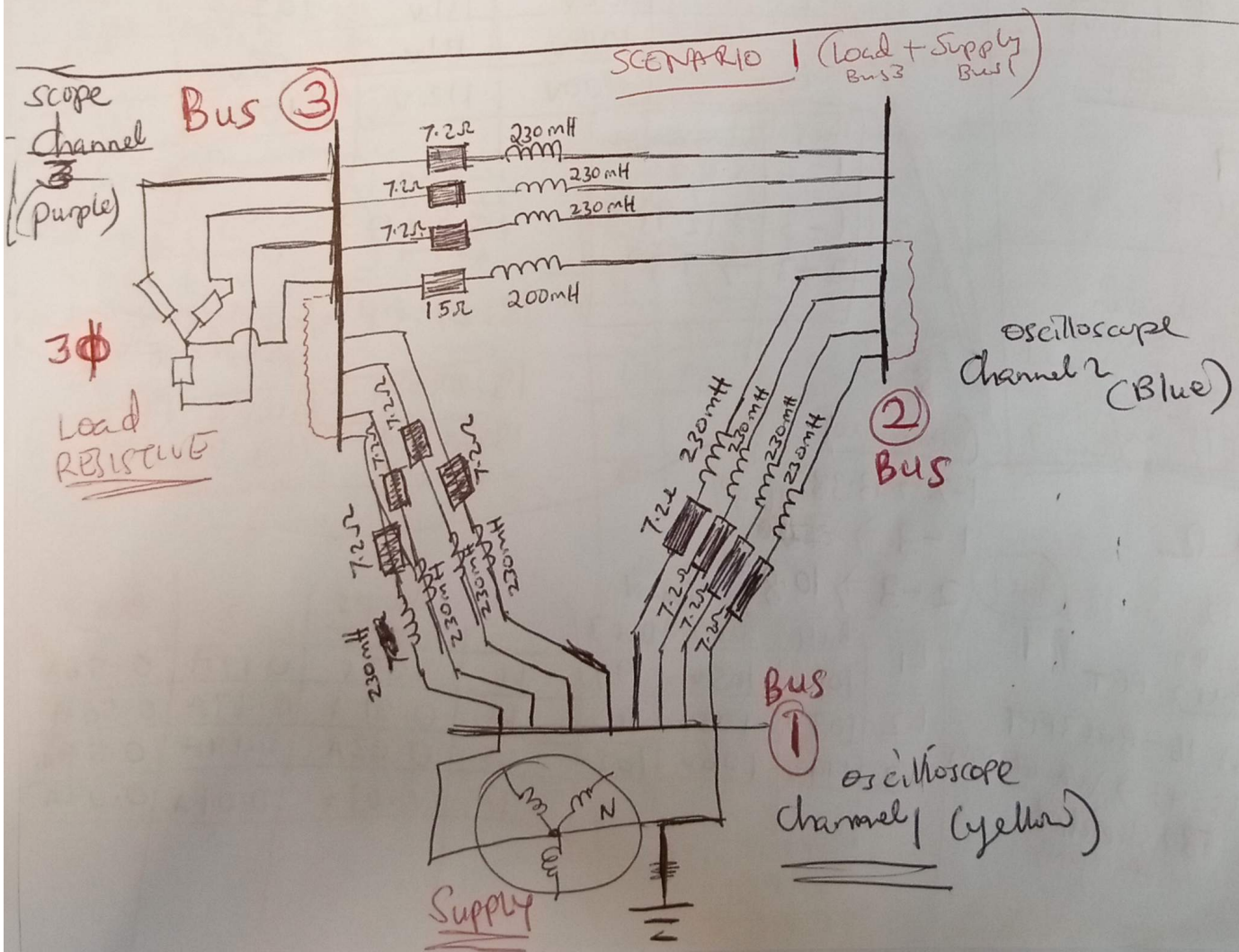
* 30 μ F $\rightarrow$ 3 ϕ

3:25 AM 6th / April / 2026 $\rightarrow$ video $\rightarrow$ iPhone

Input 3 ϕ $\Rightarrow$ 3.3V RMS Square wave *

4:03 AM - USB $\rightarrow$ DYN5 TX
 Signal $\rightarrow$ ESP32 - Input Y side

PST $\rightarrow$ 4:16 - 1 phase video



6TH/APRIL/2026

Load Bus 3

750Ω

	Supply Bus 1 Voltage (V _{pn-N})	Bus 2	LOAD Bus 3
L1	103v	112v	109v
L2	105v	113v	110v
L3	100v	114v	113v
Amps			
L1	0.45A	0.09A	0.15A
L2	0.44A	0.09A	0.15A
L3	0.44A	0.09A	0.15A
N	0.03A	0.00A	0.00A

* Voltage angle
Bus 1-2 → 6°
1-3 → 8.6°
2-3 → 1.6°

17:45:35

Image USB

Wave form

* Supply power (Bus1)

FFT

Fundamental & Harmonic

17:53:58

17:54:26

18:07:08

18:07:33

Bus 2 FFT

18:19 → Waveform

750Ω

Time div stretched

500Ω

	Bus 1	Bus 2	Bus 3
L1	0.49A	0.12A	0.30A
L2	0.48A	0.12A	0.30A
L3	0.48A	0.12A	0.30A
N	0.03A	0.01A	0.01A

VOLTAGE PHASE TO NEUTRAL

L1	102v	111v	107v
L2	104v	111v	108v
L3	100v	112v	110v

Bus 1	mean angle
1-2	→ 8.8°
1-3	→ 12.75°
2-3	→ 3.8°

18:26:17 → 2 cycle waveform

18:27:58 - Supply power (Bus1) FFT

18:28:49 - Bus 2 FFT

18:29:42 → Bus 3 FFT

↑ Random image for channel 1

channel 2, channel 3

FFTs

200Ω

18:32 - Supply (Bus1) FFT

18:32:40 Bus 2 FFT

18:33:16 - Bus 3 FFT

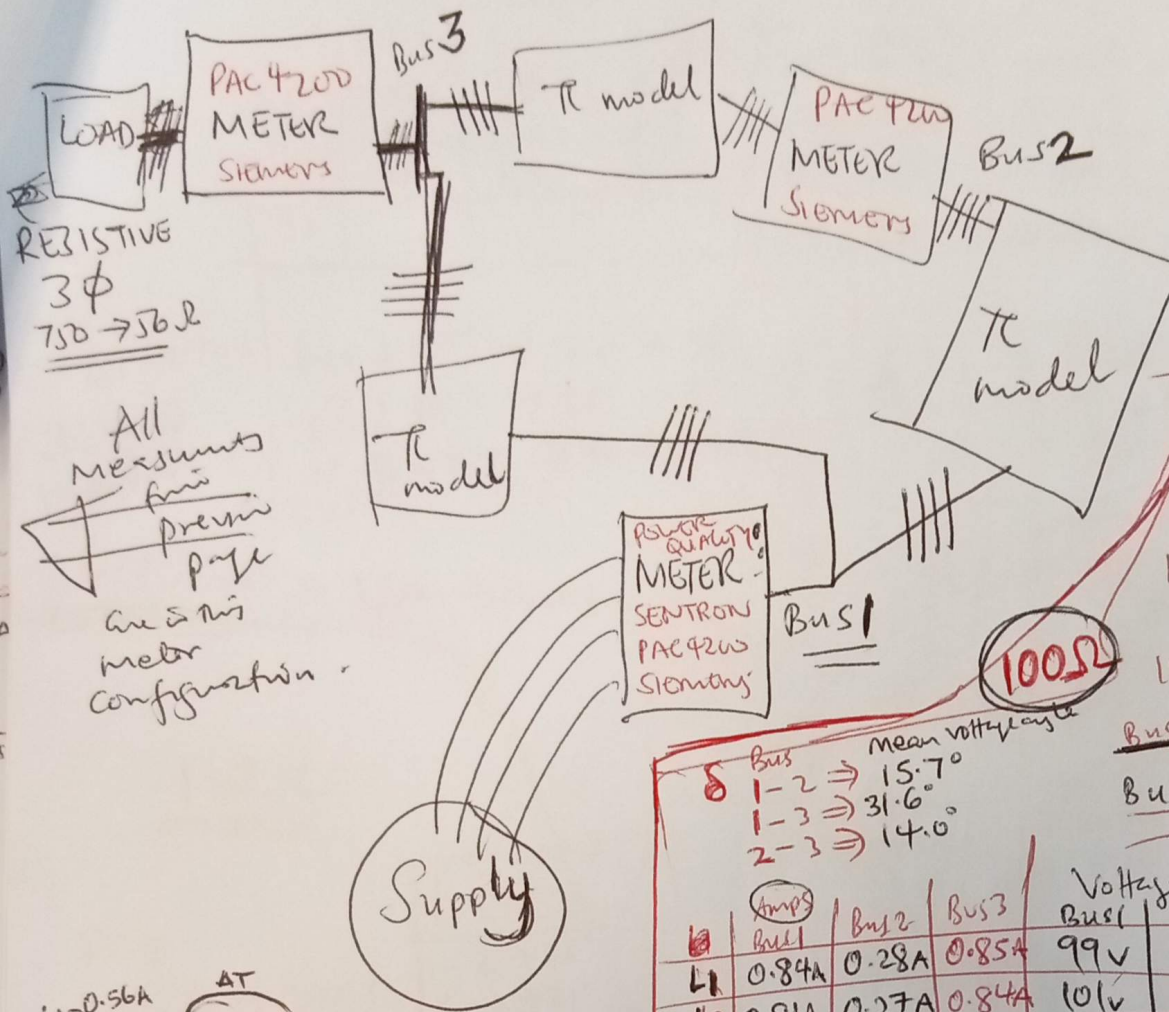
18:33:43 → WAVEFORM

18:33:58 → stretched waveform

(Voltage angle)
1-2 → 13.38° 21.5°
1-3 → 21.5°
2-3 → 10.8°

V_{pn-N}

	Bus 1	Bus 2	Bus 3	Amps	Bus 1	Bus 2	Bus 3
L1	101v	105v	99v	L1	0.63A	0.19A	0.56A
L2	103v	105v	100v	L2	0.61A	0.19A	0.56A
L3	99v	106v	102v	L3	0.62A	0.19A	0.54A
				N	0.03A	0.01A	0.02A



6th/Apr/2026

19:37:01 - Sketched waveform
19:36:41 waveform

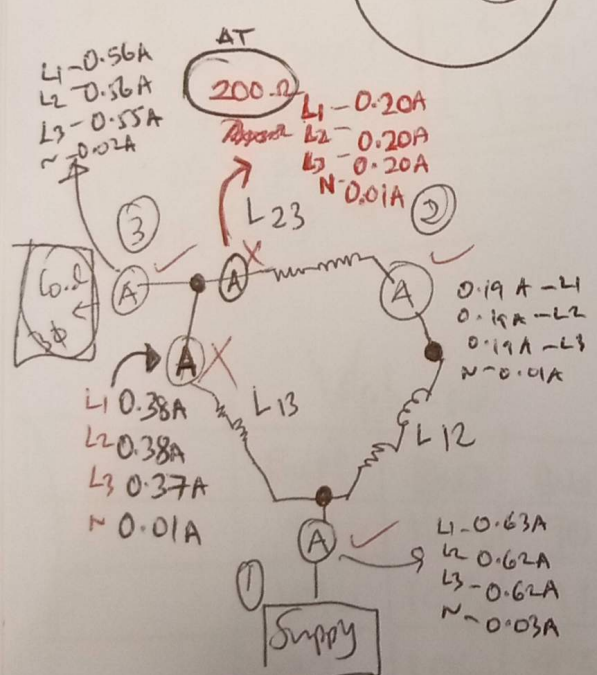
19:38:31 - FFT Bus 3

Bus 2 $\rightarrow$ 19:39:40

Bus 1 $\rightarrow$ 19:41:00

Mean voltage angle
Bus 1-2 $\Rightarrow$ 15.7°
1-3 $\Rightarrow$ 31.6°
2-3 $\Rightarrow$ 14.0°

	Amps Bus 1	Bus 2	Bus 3	Voltage Bus 1	Bus 2	Bus 3
L1	0.84A	0.28A	0.85A	99v	96v	87v
L2	0.81A	0.27A	0.84A	101v	97v	90v
L3	0.82A	0.27A	0.83A	97v	97v	91v
N	0.02A	0.01A	0.03A			



56 Ω

	Vph-n Bus 1	Bus 2	Bus 3	Amps Bus 1	Bus 2	Bus 3
L1	98	81	67	1.18A	0.42A	1.28A
L2	99	82	68	1.14A	0.41A	1.27A
L3	96	82	70	1.16A	0.40A	1.25A
N				0.03A	0.02A	0.04A

20:02:51 - waveform / 20:07:30

20:05:49 - Sketched waveform

FFT Bus 1 - 20:07

FFT Bus 2 $\rightarrow$ 20:07:52 and 20:08:26

FFT Bus 3 $\rightarrow$ 20:08:50

1-2 $\rightarrow$ 19°
1-3 $\rightarrow$ 48°
2-3 $\rightarrow$ 28°

SERIES COMPENSATION AT BUS 2

30nF

VOLTAGE

AMP'S

	Bus1	Bus2	Bus3	Bus1	Bus2	Bus3
L1	102v	116v	104v	0.49A	0.18A	0.25A
L2	105	116v	105v	0.47A	0.18A	0.28A
L3	101	117v	107v	0.48A	0.18A	0.28A
N				0.03A	0.01A	0.01A

500Ω

Bus Configuration

674/14/26
20:26:55
Waveform

Bus 3 FFT → 20:30:36
Bus 2 FFT → 20:30:55
Bus 1 FFT → 20:32:27

δ
1-2 → 9.6°
1-3 → 12.4°
2-3 → 20.5°

L1	0.15A
L2	0.15A
L3	0.15A
N	0.01A

L1	0.18A
L2	0.18A
L3	0.18A
N	0.01A

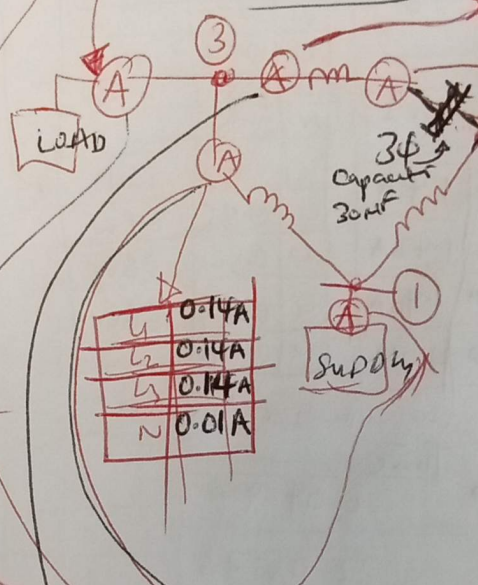
SETUP → 20:58pm → iPhone

100Ω

δ (voltage angle)	
1-2	+25°
1-3	+21°
2-3	-2.5°

FFT

21:13 → Bus 1
21:14:07 ← Bus 2
21:14:26 ← Bus 3
21:14:45 → Waveform



	Watt
L1	0.37A 33w
L2	0.37A 34w
L3	0.36A 33w
N	0.02A

L1	0.48	44
L2	0.49A	45
L3	0.49A	46
N	0.04A	

V_{ph-N}

AMPS

Bus1	Bus2	Bus3		Bus1	Bus2	Bus3
99v	106v	90v	L1	0.87A	0.48A	0.85A
101v	105v	91v	L2	0.86A	0.48A	0.86A
97v	106v	93v	L3	0.87A	0.48A	0.85A
			N	0.04A	0.04A	0.03A

Watts

L1	85	47	76w
L2	86	48	78w
L3	84	48	79w

Compensate & ~~node~~ node

56Ω

	Watts	
L1	101w	1.42A
L2	109w	1.43A
L3	109w	1.40A
N		0.03A

Normal Compensation
As previous

V_{d-n} Bus1 Bus2 Bus3

	Bus1	Bus2	Bus3
L1	97v	95v	75v
L2	98v	94v	76v
L3	95v	96v	78v

Amp 8

	Bus1	Bus2	Bus3
L1	✓		
L2	1.36A	0.82A	1.41A
L3	1.33A	0.82A	1.42A
N	1.35A	0.82A	1.40A
	0.05A	0.07A	0.03A

Doesn't change with meters re-shuffling to the location.

Bus	FFT → Time
Bus1 FFT	22:04:44
Bus2 FFT	22:05:23
Bus3 FFT	22:06:05

Waveform

(V_s, V_R) All three Buses → 22:02:30 Am → 6th Apr 2026
22:03:09 Am - started waveform.

(Voltage angle)	Bus no	Degrees
	1 → 2	45.40
	1 → 3	33.50
	2 → 3	-7.50

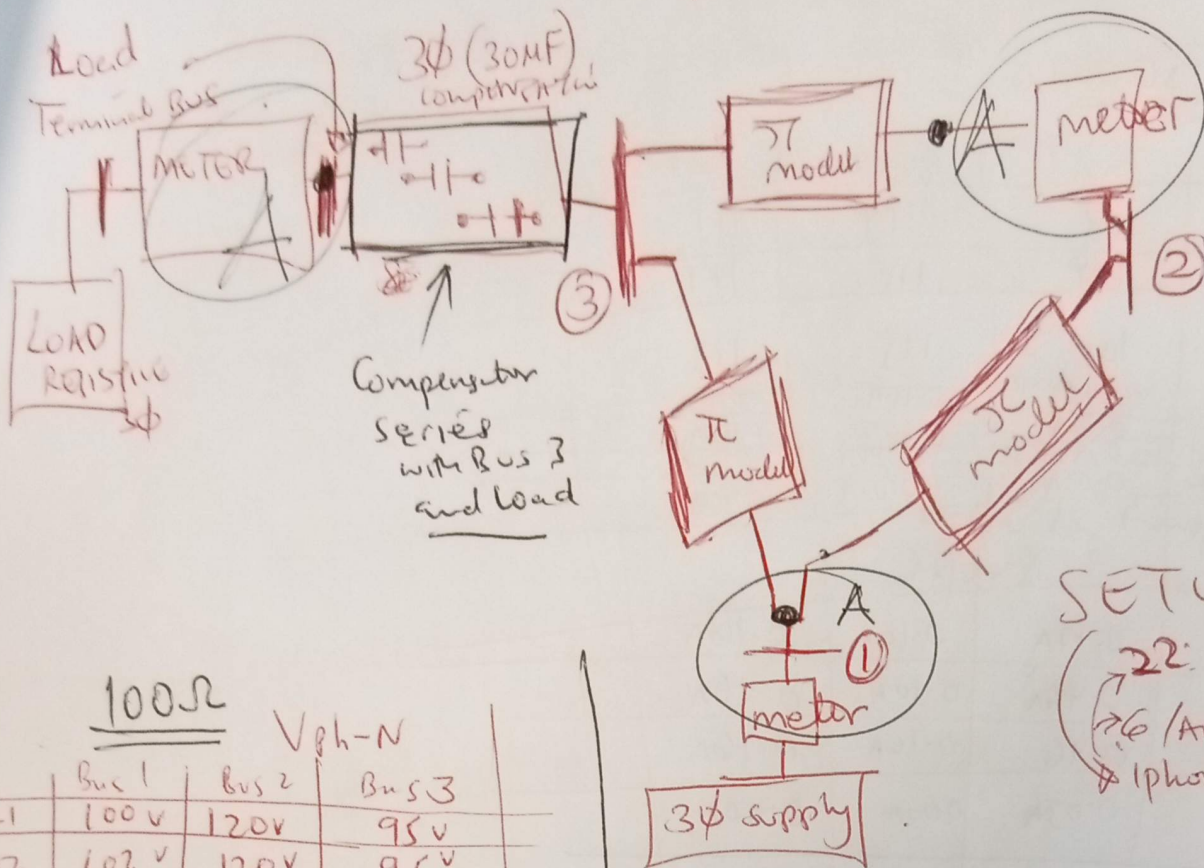
Wd
3rd
Resistance

System

	Amps
L1	0.82A
L2	0.82A
L3	0.82A
N	0.07A

600

$$F = 49.99112 \sim 50 \text{ Hz}$$



SETUP
(22:47 pm
6 April 2026
iPhone

<u>100Ω</u> V_{gh-N}			
	Bus 1	Bus 2	Bus 3
L1	100V	120V	95V
L2	102V	120V	95V
L3	99V	122V	97V
<u>Ampl</u>			
	Bus 1	Bus 2	Bus 3
L1	1.19A	0.35A	0.89A
L2	1.16A	0.35A	0.89A
L3	1.19A	0.35A	0.88A
N	0.04A	0.01A	0.04A

Waveform $\rightarrow$ 22:46:21 USB

8-1-2 $\rightarrow$ 15-40

$$1-3 \Rightarrow -14.5$$
$$2-3 \Rightarrow -29.26^\circ$$

2251:01 → FFT Bus 3

22:54:41 ← PFT-Boss2

22:55:28 ← FFT BUS 1

	56Ω		
	BUS 1	BUS 2	BUS 3
L1	1.83A	0.54A	1.40A
L2	1.76A	0.53A	1.40A
L3	1.78A	0.51A	1.36A
N	0.07A	0.02A	0.07A
	Vph-m		
L1	99V	133V	74V
L2	100V	133V	75V
L3	97V	136V	75V

FFT $\rightarrow$ Busi $\rightarrow 22:53:53$.

F&T → Bus 2 → 23:00:41

Art $\rightarrow$ Bus 3 $\rightarrow$ 23:01-15

23:01:39 → WAVEFORM

~~✱~~ BELL LABS / SPACE
/ NASA LAB
SETUPS / EXPERIMENT
DESIGNS
↓
~~✱~~ Adopt in your research.

750 R

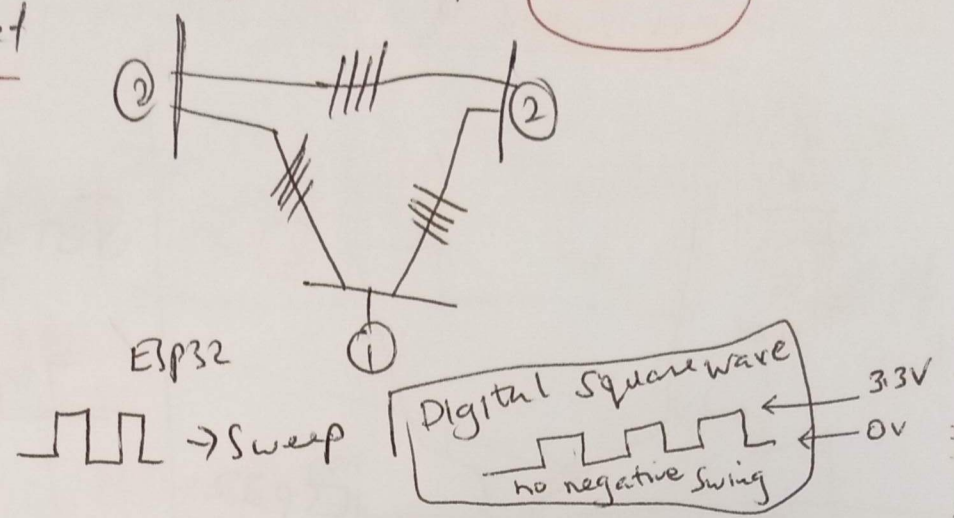
	Bus 1	Bus 2	Bus 3
L1	103	114	111
L2	104	115	111
L3	101	115	113
Amps			
L1	0.47A	0.10A	0.16A
L2	0.46A	0.10A	0.16A
L3	0.46A	0.09A	0.16A
N	0.03A	0.00A	0.00A

Waveform → 23:03:27

Amps

L1	0.47A	0.10A	0.16A
L2	0.46A	0.10A	0.16A
L3	0.46	0.10A	0.16A
N	0.03A	0.00A	0.00A

23:53 pm → Image of setup - iPhone SETUP
 * Uncompensated



Bus 1 → Signal (Square wave)
 3φ

Measurement at Bus 3 *

50Hz → FFT → 0:11:20 - 7th April 2026
 ← 0:12:01

Waveform
 21Hz FFT
 Fundamental
 ← 0:17:11 - 7th/Apr/2026
 ← 0:18:40
 WAVEFORM

300Hz - waveform → 0:22:01

FFT → 0:25:20

0:31:19 - waveform of a resonant point?
 FFT? → 150Hz fundamental → 00:38:50

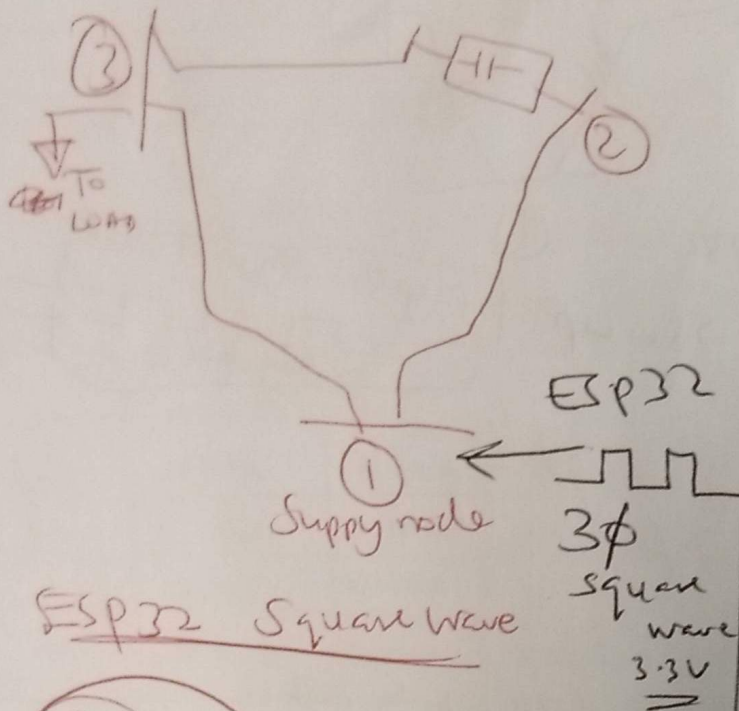
~~0:33:08~~
 0:37:25 combined FFT 2 waveforms.

Bus 2

00:48:44 - Combined FFT & waveform
 00:49:35 - waveform
00:51:50 * FFT

Resonant
 (Amplitude
 greater than ~~rest~~
 the rest of
 frequencies.)

Series compensated node



ESP32 Square wave

Bus 2

Bus 2 - Resonance point

* Waveform + FFT → 1:12:10
 * Waveform only ← 1:12:39
 FFT ← 1:13:24

7th April 2026

Bus 3

1:16:26 - 300Hz waveform
 1:18:24 - FFT

Waveform 1:23:47
 Video FFT 1:25:47
 iPhone

FFT → 1:20:25 → 150Hz → further
 1:21:08 - waveform

April 23rd / 2026

